NHID-Clinical

ENGINEERING REFERENCE

v2 Technical Playbook

NHID-Clinical · Tech Stack, Adoption Tiers & Governance Framework Reference

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OPEN PROPOSAL — NOT A STANDARD OR REGULATORY REQUIREMENT

NHID-Clinical is an early-stage open proposal by Brianna Baynard. It is a voluntary open proposal — not an accredited standard, not a certification program, and not a regulatory requirement. No HIPAA compliance, security guarantees, or liability protection are implied.

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Core Objectives

• Establish a voluntary behavioral baseline — disclosure, data-gating, escalation, and audit — for AI voice agents in B2B healthcare payer–provider calls.

• Provide an open, testable reference: every claim is backed by runnable, deterministic code.

• Layer in cryptographic agent identity (NHID-Auth v2) as authorization matures beyond behavior alone — NPI-bound passports, scoped delegation, per-agent revocation.

• Make Call Authorization Score (CAS) a procurement-grade compliance signal payers and vendors can both verify independently.

Five-Layer Trust Stack

Layer 0 · NPI Gap
The problem — no cross-org NPI auth

Layer 1 · STIR/SHAKEN
RFC 8224 — carrier number auth (A/B/C attestation)

Layer 2 · NHID-Clinical v1.3
Behavioral baseline — 5 controls, 18 CTS tests

Layer 3 · NHID-Auth v2
Cryptographic agent identity — Ed25519, NPI binding

Layer 4 · FHIR / OpenTelemetry
Healthcare-native audit + SIEM export

Tech Stack & Repository Layout

Layer	Component	Purpose
Policy engine	src/nhid_policy_engine_v1.py	Deterministic rule evaluation (670+ lines)
Identity (v2)	src/agent_identity.py	Ed25519 delegation & agent passports
Scoring	src/nhid_cas.py	Call Authorization Score engine
Audit	src/fhir_audit_emitter.py	FHIR R4 AuditEvent generation
Conformance	tests/nhid_conformance_test_suite_v1.yaml	18 deterministic CTS cases
Adapters	adapters/*.py	VAPI, Twilio, Vonage, Retell, Amazon Connect, call-progress
Compute	functions/handler.py	AWS Lambda entry point
Deploy	template.yaml	AWS SAM CloudFormation

Integration Ladder — Tier 0 to Tier 2

Each tier is independently useful. Stop at any rung — v2's cryptographic layer is powerful but it is not the on-ramp.

Tier	Time	What You Get	What You Need
0	15 min	Conformance verdict + CAS score per call	A transcript and curl
1	~2 hr	Automated per-call checks in your pipeline	An end-of-call webhook
2	~1 day	Cryptographic agent identity (NPI-bound)	pip install cryptography

Governance Framework — Call Authorization Score (CAS)

CAS provides a continuous compliance signal between 0.0 and 1.0 per call session: CAS = F_IAF × F_NOCF × ECF — Identity Assurance, Operational Conformance, and Evidence Completeness factors.

CAS Score	Tier	Badge
≥ 0.90	Verified Trust	L2
≥ 0.75	Conditional Trust	L1
≥ 0.50	Review Required	—
≥ 0.20	Denied / Degraded	—
< 0.20	Hard Denial	—

Policy Engine Action Priority

Priority	Action	Trigger
5	DENY_DATA	IDG-01 or PDX-01 critical violation
4	ESCALATE_HUMAN	EIT-01: escalation requested, path available
3	DISCLOSE_IDENTITY	IDG-01: no prior disclosure detected
2	LOG_ONLY	DBC-01 text heuristic (non-blocking), ATR-01 minor gap
1	CONTINUE_AI	All controls pass

Live API Quick Reference

Method	Endpoint	Auth	Purpose
POST	/v1/demo/check	none	Raw NHID event → result
POST	/v1/adapters/vapi/check	none	Native VAPI payload → result
POST	/v1/adapters/twilio/check	none	Native Twilio payload → result
POST	/v1/conformance/check	**x-api-key**	Production conformance check
GET	/health	none	Liveness probe

Base URL

https://gfvq4swdtf.execute-api.us-east-1.amazonaws.com/prod